MBA ZG583 · MANAGEMENT OF AI PRODUCTS · CAPSTONE

UniConnect Operations OS

An AI-assisted academic operations platform running across a 40-campus higher-education network. Product review for a pilot decision.

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01 The same operational day, forty times over

Fees to reconcile, daily reports to write and review, attendance to record, problems to route — every working day, at every campus. Before the platform it lived in spreadsheets, email and WhatsApp. The platform fixed the structure. What remains is a reading problem.

40

campuses filing a free-text report every working day

2.5 hrs

between the 16:00 filing deadline and the 18:30 sign-off

1–2 days

typical lag before a serious issue is escalated

The failure is not a missing feature. It is that decision quality falls as the network grows, because the reading does not scale.

02 Who feels it, and when

Business Operations Associate

- Files one campus report before 16:00
- Wants it accepted first time
- Re-sends when it is sent back

Programme Manager / COS

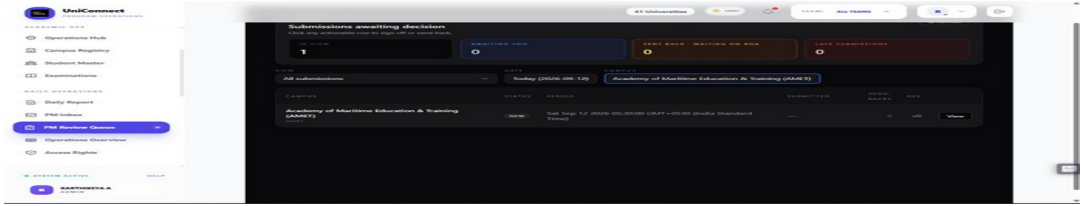
- Reviews ten to twelve campuses nightly
- Must find the three that need tonight
- Signs off or escalates

Academic Operations Lead

- Plans exam cycles per campus
- Portions, papers, seating, invigilation
- Rebuilt in spreadsheets today

The reviewer's evening is the moment the product is designed around

submit → 16:00 deadline → **AI-assisted triage** → **human review** → sign off / send back / escalate → immutable audit



The review surface today: submissions awaiting a decision, in arrival order, with no ranking applied.

03 Why AI, and why not

Rules and dashboards already handle this

- A fee number that drops
- Attendance below target
- A report not filed by 16:00
- Any numeric threshold breach

They cannot handle this

- "lab machines down again since Monday"
- Buried in paragraph four of a routine report
- Written by forty different people
- A language problem, not a numbers problem

Why now

- The network reached 40 campuses — the reading load crossed what a small central team can absorb.
- The platform now supplies both things AI needs here: structured context to ground on, and an existing human approval step to catch it when wrong.
- Inference is a per-call cost in fractions of a rupee, so the entry cost is near zero rather than a licence commitment.

Where AI is deliberately absent: fee arithmetic, permission enforcement, approval states and audit records stay deterministic — they must be exactly correct and defensible to an auditor.

04 Platform, plus one narrow wedge

PLATFORM — live in production

One database, one permission model, one audit trail. Fees · daily operations spine · tasks · campaigns · assessments and exam papers · faculty · budgets · meetings · academic operations hub

INTELLIGENCE LAYER — a feature, not a second product

Reads the operational text nobody has time to read, states how confident it is, and routes each item to the right human with the evidence attached. Switch it off and the platform still runs.

MVP — five campuses, one cluster, one academic month, assist-only

In scope

- Summary with cited source sentence
- Severity and confidence ranking
- Override with structured reason
- Weekly labelled evaluation sample

Out of scope

- Any automatic approval
- Any student-facing message
- Predictive scoring of a named person
- Model write access to operational tables

05 The decisions worth defending

Layer	Decision	Why
Model / inference	API — Google Gemini	Quality is adequate, cost is per-call, switching cost stays in our code
Workflow, permissions, audit	Build	The differentiation, and the part that must be exactly correct
Database, hosting, queues	Buy — Supabase, Railway, Redis	Row-level security at the database layer for a fraction of the cost
Identity, mail, calendar	Partner — Google Workspace	The institution already runs on it

Deferred: predictive student-risk scoring

The training data reflects recording habits more than student behaviour, and a wrong prediction attaches to a named student.

Rejected: AI-scored staff performance

It turns an operations tool into a surveillance instrument, which destroys the honest daily reporting the platform depends on.

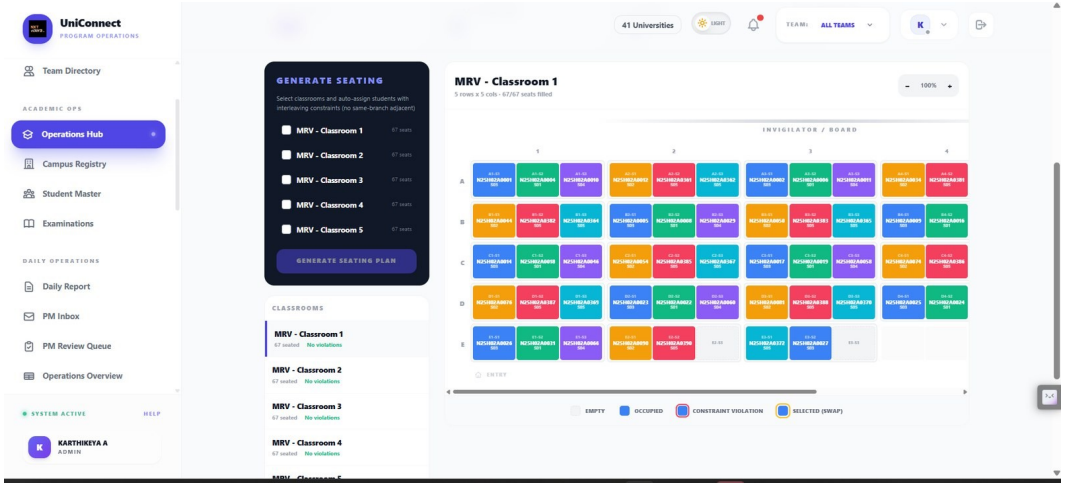
06 Phases end at proof gates, not at dates

Phase 2 does not start because Phase 1 finished. It starts because Phase 1 met a measured threshold.

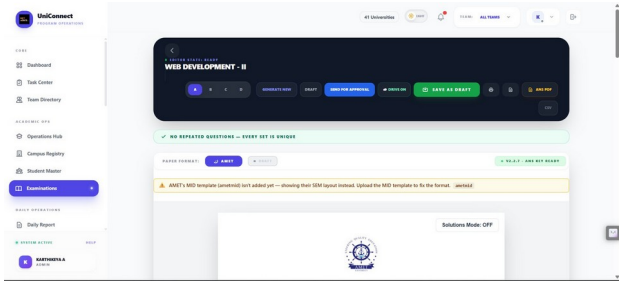
Phase	Focus	Gate before proceeding
0 — done	Deterministic platform; narrative summaries and meeting intelligence live	Adopted as the daily system of record
1 — MVP pilot	Daily operations triage on five campuses for one month, assist-only, with the evaluation harness	Recall, groundedness, override band and reviewer trust all met
2 — expansion	All 40 campuses; cross-campus patterns; OCR intake; budget completeness checks	Cost per campus-month within ceiling; zero scoping breaches
3 — assisted exams	Paper equivalence assistance and portion inference, faculty approval mandatory	Zero AI-only paper approvals; academic governance sign-off
4 — productisation	Assess offering the platform to peer institutions	Two stable quarters and a defensible support model

Two AI ideas were cut from this plan rather than sequenced later: predictive student-risk scoring and AI-scored staff performance. Cost of error, not value, is what removed them.

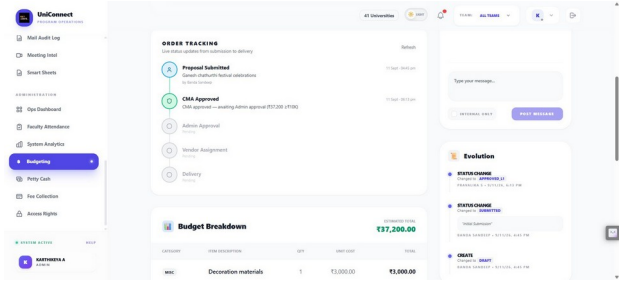
07 The prototype is the production system



Seating allocation: 67 of 67 seats filled, interleaving constraint satisfied, every classroom reporting no violations. An invalid plan is flagged, never silently emitted.



Paper generation ends at Send for Approval — a person approves, and any question can be swapped.



The budget path: no AI anywhere in it, threshold rule printed on the screen.

08 How we would know it worked

Two baselines run alongside the pilot, so the comparison is measured rather than argued: today's chronological queue, and a rules-only ordering that could be built in a week.

Gate	Threshold, fixed before the pilot
Serious-issue recall on the weekly labelled sample	≥ 0.90 sustained over four weeks
Summary statements not supported by the source	≤ 2% of sampled statements
Reviewer override rate	Between 10% and 25% — two-sided
Campus-scoping or permission violations	Exactly zero — absolute blocker

The override band is deliberately two-sided. Below 10% is not success — it is the first symptom of reviewers rubber-stamping, which is the failure this design fears most.

09 Economics: one campus, one month

Inference and infrastructure per campus, plus amortised maintenance, must stay materially below the loaded cost of the reviewer and associate hours removed — at every campus count between five and forty.

Why it should hold

- A daily report is a small amount of text
- ~22 working days bounds call volume
- The hours displaced are managerial

What could break it

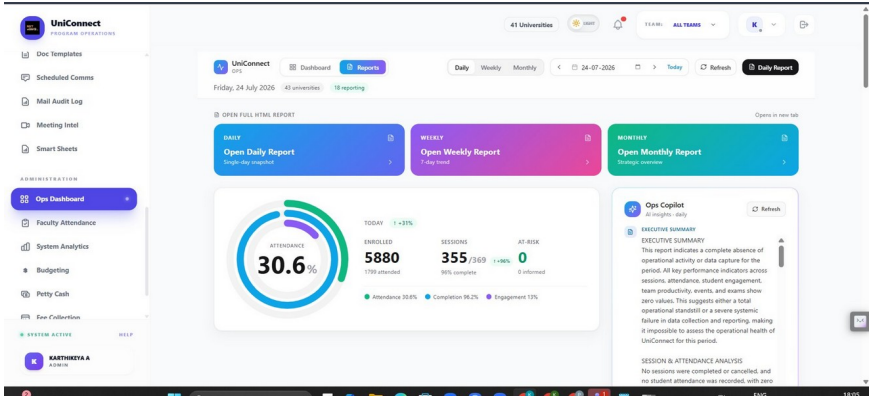
- Context size growing beyond seven days
- Extending to high-volume, low-value items
- Both controlled by pinned configuration

Rupee figures are computed from measured pilot volumes and current API pricing rather than presented as an estimate. A fabricated cost table would be the fastest way to lose the argument.

Scaling risks: latency at the 16:00 burst · linear cost growth · per-campus exception creep · governance overhead · single-builder dependency

10 The risk that actually worries us

Automation complacency: reviewers stop reading the source reports, oversight becomes ceremonial, and the platform has made the institution less informed while appearing more informed.



Risk R1, demonstrated rather than asserted: on this day the generated summary reported a complete absence of activity, on a screen showing 355 of 369 sessions completed.

Controls that follow from it

- AI cannot advance a state or send a message
- Every generated claim cites its source sentence
- Rules can raise severity, never lower it
- Auto-sign-off disabled on serious items
- Override reasons read by a human weekly
- Row-level security at the database, not the app
- Identifiers stripped before the API call

RECOMMENDATION

Pilot. Five campuses, one academic month, assist-only.

Scale the deterministic platform across all 40 campuses in parallel. Do not extend AI into exam approval, student-facing communication or staff evaluation until the gates are met.

We expand when

- Recall holds at 0.90 for four weeks
- Ungrounded statements stay under 2%
- Override rate sits inside 10–25%
- Reviewers say the queue changed their order

We stop when

- Any cross-campus data exposure, even once
- Ungrounded statements above 5%
- Override rate below 10% — rubber-stamping
- Anyone proposes using it to rate staff

A team that can say precisely what would make them stop is a team worth funding.